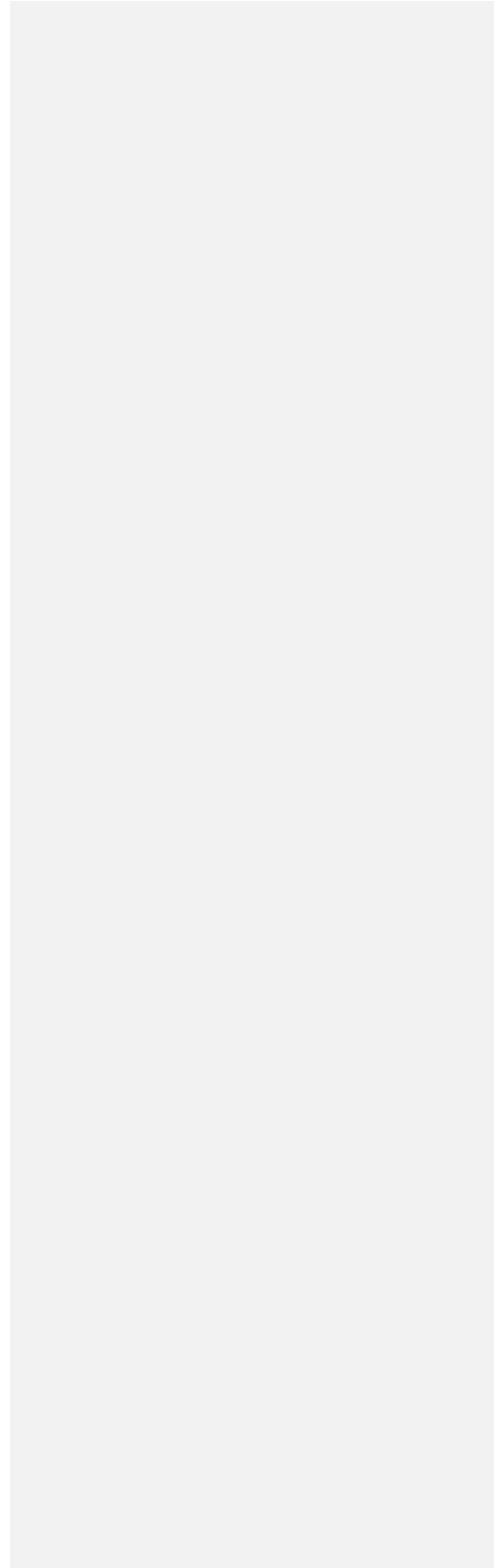


◆ Beta Project charter



PROJECT INITIATION DOCUMENT (PID)

Project Name: Implementation of Industry 4.0 Digital Transformation

Organization: Beta

Date: July 1, 2026

Reference:

1. PROJECT CHARTER

1.1 Purpose & Justification

This project is initiated following approval from the Ministry of Industry to implement the outcomes of the Digital Transformation Roadmap (Fourth Industrial Revolution). The primary justification is to modernize manufacturing capabilities by integrating advanced digital solutions onto existing machinery, thereby enhancing operational efficiency and monitoring capabilities.

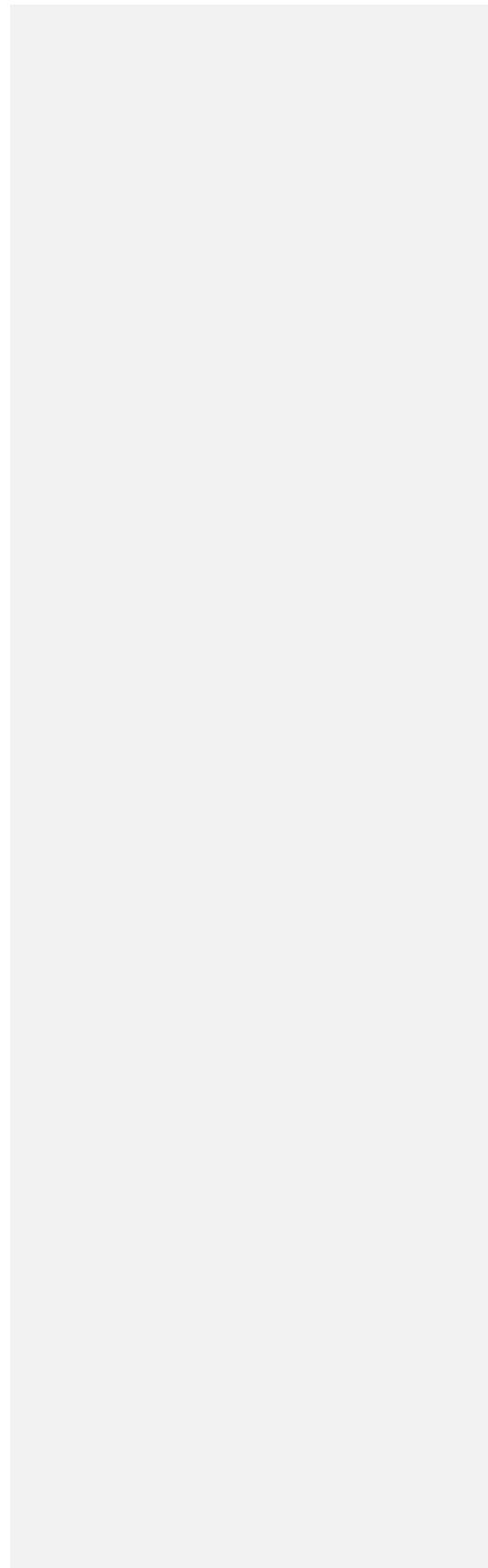
1.2 Project Objectives

- Monitor production activities accurately through digital systems.
- Significantly increase overall operational efficiency.

1.3 Project Scope

- **Contracting:** Engaging a specialized technical company for solution implementation.
- **Implementation:** Supply and installation of digital systems onto production machines.
- **Supervision:** Direct oversight by the Ministry of Industry and an internal committee.

Beta Requirements



Software Requirements Specification: EdgeView Industrial Platform

1. Introduction

The system is an Industrial IoT (IIoT) and Manufacturing Execution System (MES) designed to monitor machine health, manage work orders, track energy consumption, and calculate Overall Equipment Effectiveness (OEE).

1.1 Scope

The system encompasses four primary modules:

1. **Work Order Management**
2. **Machine Efficiency (OEE)**
3. **Power Monitoring**
4. **Financial & Reporting**

2. Functional Requirements

2.1 Work Order Management Module

Objective: Manage production tasks from assignment to completion, ensuring full traceability.

- **Work Order (WO) Lifecycle:**
 - The system shall support the following WO statuses: **Ready, In Progress “Open, On Hold”, Finished.**
 - The system shall allow the assignment of WOs to specific machines through Production Admin Credentials .
 - Using appropriate credentials to link wo with machine & operator
 - System shall have product traceability along different WOs.
- **Data Inputs:**
 - **Master Data:** Refer to process flow page 12.
 - **Production Data:** Start & Finish times for each WO & wo outputs .
- **Work Order Dashboard:**
 - **KPI Cards:** Display counts for Total Work Orders, Ready, In Progress, and Finished.
 - **Production Metrics:**
 - Display Total Outputs (units)
 - Total Output Weight (Kg)
 - Avg Completion Time (Hours)
 - Avg Output Weight.
 - **Visualizations:**
 - Status Distribution (Donut Chart).
 - Daily Production (Last 7 Days) comparing Completed Orders vs Output Weight.
 - Hourly Production Distribution (Bar Chart).
 - Production by Machine Type (Pie Chart).

2.2 Machine Efficiency & OEE Module

Objective: Monitor real-time machine performance and historical trends.

- **Machine Types Supported:**
 - Armor Rod & Guy Grip Production Line:
 - **Forming Machine (Phase 1)**
 - Round Head Machine
 - Gluing and Sanding Machine
 - Twisting Machine
 - Bending Machine
 - **Flat Washer Line (Phase 1)**
 - Spring Washer Line
 - Bolts Production Line
 - Nuts Production Line
- **Dashboard & Monitoring:**
 - **Overview Dashboard:** Display a high-level card for each machine showing:
 - Status Tag (e.g., "Running", "Stopped").
 - Live OEE %, Live Power (kW).
 - Visuals for Availability, Performance, and Quality.
 - Units produced (Total vs. Good).
 - **Detailed Machine Page:**
 - **Live KPI Cards:** Current OEE, Current Power (kW), Production (24h), Quality Rate.
 - **OEE Breakdown:** Separate progress bars for Availability, Performance, and Quality.
 - **Statistics:** Uptime counters, Downtime counters, Energy consumed (kWh).
 - **Trend Charts:** Line charts for OEE trends over the last 24 hours (plotting OEE, Availability, Performance, and Quality).
- **Data Acquisition (DAQ) Requirements:**
 - The system must extract the following **Production Data** from all machine types:
 - Machine Total Input (Meter)
 - Machine Throughput (Kg)
 - Run & Stop Status
 - New Output (Piece) & New Output Weight
 - **Manual Inputs:** The system shall allow operators to manually input
 - **Downtime Reasons.**
 - Stops definitions and down Time Pareto

2.3 Power Monitoring Module

Objective: Track energy usage to identify costs and deviations.

- **Electrical Parameters:**
 - The system must monitor the following for every machine:
 - Consumed Energy (kWh)
 - Power (kW) & Peak Power (kW)
 - Line Voltages (V) & Phase Voltages (V)
 - Line Current (A)
 - Frequency (Hz)
 - Power Factor
 - Phase Unbalance
- **Power Dashboard:**
 - **KPIs:** Total Energy (kWh), Average Power Draw (kW), Peak Power.
 - **Visualization:** A multi-line trend chart comparing energy consumption across different machines (e.g., Cutting 1 vs Extruder 1 vs Printing 1).
 - **Machine Cards:** Individual cards displaying live Current (kW), Today's Energy (kWh), Average Voltage, and Average Current per machine.

2.4 Analytics & Reporting Module

Objective: Provide historical analysis and exportable reports.

- **Filtering:** Users must be able to filter data by **Date Range** (From/To) and **Machine** selection.
- **Analysis:**
 - **Trend Graph:** Historical OEE performance line chart.
 - **Performance Summary Table:** A grid view displaying:
 - Machine Name & Code
 - Avg OEE %
 - Availability / Performance / Quality %
 - Total Production / Good Units / Defects / Defect Rate
 - Uptime / Downtime durations.
- **Export Options:** The system shall support exporting reports to **Excel** and **PDF**.

2.5 Financial Module (Business Output)

- **Cost Analysis:** The system shall utilize energy monitoring data to calculate the **Actual Cost of Energy** for the factory.
 - **Revenue Monitoring:** The system shall use traceability features to monitor **Monetized Revenue**.
 - **Loss Analysis:** The system shall identify the **Cost of Poor Quality** based on scrap rates and reject logging.
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3. Data Flow & Integration

Source: Flowchart "Final Product Workorder"

The system shall be able to integrate with client ERP "Odoo" to extract master Data and Send Production Data.

**In phase 1 the data will be manually entered until we integrate with Odoo ERP*

1. **Downstream Flow (Master Data):**
 2. **Upstream Flow (Production Data):**
 - Machine Metrics (Power, OEE, Traceability) are captured at the machine level.
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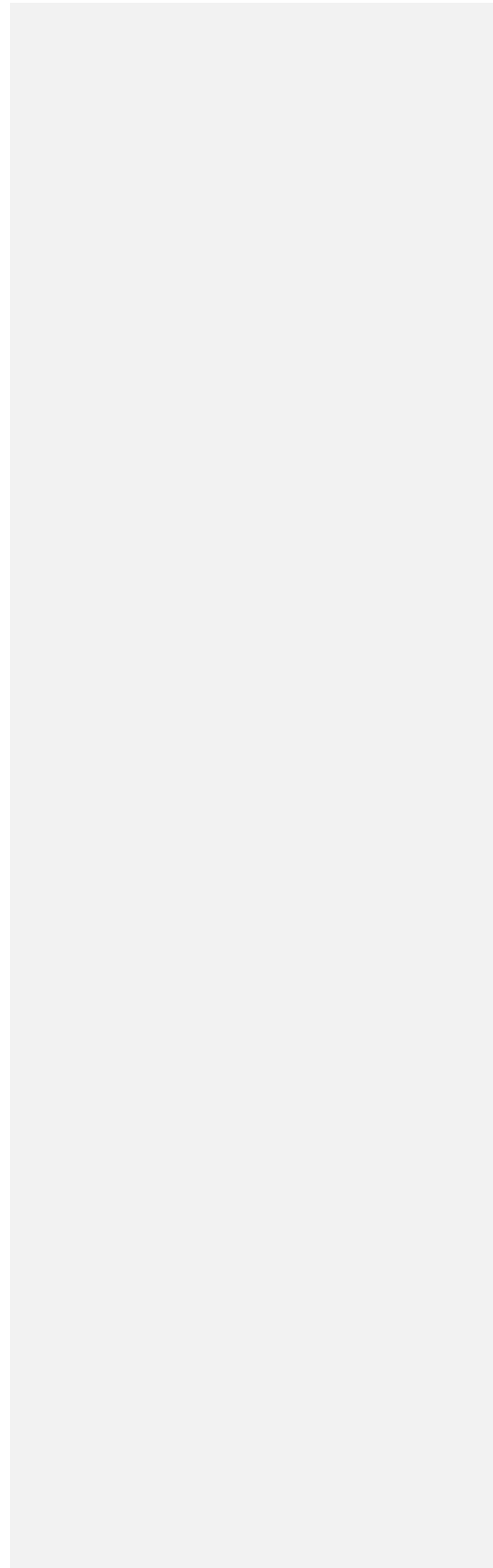
4. User Interface (UI) Structure

Source: Dashboards Outline & Screenshots

The application shall consist of a sidebar navigation with the following hierarchy:

1. **Dashboard** (Global Overview)
2. **Machines**
 - List view / Card view of all machines.
 - Drill-down to "Detailed Machine Parameters Live".
3. **OEE Reports** (Analytics interface)
4. **Power Monitoring** (Energy specific dashboard)
5. **Work Orders**
 - Overview Dashboard
 - Detailed Workorder Page
 - Management List

Equations and calculations



factory dashboard

AVG OEE: factory Behaviour/Mo

Total energy: factory consumption
How much I paid last Month

units produced: How much I produced
ROI

Finished w.o.s: factory Behaviour/Mo

Availability: Machine health

Performance: ~~How~~ factory performance
& planning validation

Quality: Rate of loss & scrap

- **FDS-CALC-OEE:** Standard Industry 4.0 calculation:

- $Availability = \frac{Run_Time}{Planned_Time}$
- $Performance = \frac{Total_Units \times Ideal_Cycle_Time}{Run_Time}$
- $Quality = \frac{Good_Units}{Total_Units}$

1. Machine Efficiency & OEE Module

This module relies on precise timing and counting.

A. Time-Based Metrics

- **Planned Production Time ($T_{planned}$):**
 - **Source:** User Input (Shift Schedule).
 - **Equation:**

$$T_{planned} = \text{Shift Duration} - \text{Planned Breaks}$$

- **Run Time (T_{run}):**
 - **Source:** Calculated.
 - **Equation:**

$$T_{run} = \sum (\text{Duration where Machine Status} = \text{"Running"})$$

- **Stop Time / Downtime (T_{stop}):**
 - **Source:** Calculated.
 - **Equation:**

$$T_{stop} = T_{planned} - T_{run}$$

- **Availability (A):**

- Measures how often the machine is running when it is scheduled to run.
- **Equation:**

$$A(\%) = \left(\frac{T_{run}}{T_{planned}} \right) \times 100$$

- **Performance (P):**

- Measures speed consistency against the machine's ideal design speed.
- **Inputs:** `Ideal Cycle Time` (Time to make 1 unit) or `Ideal Run Rate` (Units per minute).
- **Equation:**

$$P(\%) = \left(\frac{\text{Total Count} \times \text{Ideal Cycle Time}}{T_{run}} \right) \times 100$$

- *Alternative (if using Throughput):*

$$P(\%) = \left(\frac{\text{Actual Throughput (m/min)}}{\text{Rated Throughput (m/min)}} \right) \times 100$$

- **Quality (Q):**

- Measures the proportion of good units produced.
- **Equation:**

$$Q(\%) = \left(\frac{\text{Total Count} - \text{Reject Count}}{\text{Total Count}} \right) \times 100$$

- **OEE Score:**

- **Equation:**

$$\text{OEE}(\%) = A \times P \times Q$$

- (Note: Enter A, P, and Q as decimals for this calculation, e.g., $0.85 \times 0.90 \times 0.95$, then multiply by 100).

2. Production & Work Order Module

These parameters track the physical output of the machines.

- **Total Output (Units):**
 - **Source:** Fetch (PLC Counter).
 - **Logic:** `Current_Counter_Value` - `Start_Counter_Value` (at WO start).
- **Total Output Weight (Kg):**
 - **Source:** Fetch (Scale/Load Cell).
 - **Logic:** $\sum(\text{New Output Weight})$ per batch or continuous stream integration.
- **Average Output Weight (Kg/Unit):**
 - **Equation:**

$$\text{Avg Weight} = \frac{\text{Total Output Weight}}{\text{Total Output Units}}$$

- **Average Completion Time:**
 - **Equation:**

$$\text{Avg Time} = \frac{\sum(\text{Finish Time} - \text{Start Time})}{\text{Total Finished Work Orders}}$$

4. Financial & Cost Module

These equations translate engineering data into business insights.

- **Actual Cost of Energy (C_{energy}):**

- **Input:** `Tariff_Rate` (Currency per kWh).
- **Equation:**

$$C_{energy} = (E_{kWh_End} - E_{kWh_Start}) \times \text{Tariff Rate}$$

- **Cost of Poor Quality (COPQ):**

- **Input:** `Unit_Material_Cost`.
- **Equation:**

$$\text{COPQ} = \text{Reject Count} \times \text{Unit Material Cost}$$

- **Energy Intensity (kWh per Unit):**

- Efficiency metric.
- **Equation:**

$$\text{Intensity} = \frac{\text{Energy Consumed during WO}}{\text{Total Output Units}}$$